

Alessandro Fella

AI and Machine Learning Engineer



About me

I like solving technical problems by applying what I've learned through my academic and professional path. I am passionate about creativity in all its forms and always eager to learn something new.

Areas of specialization

Data Manipulation • Data Science • Machine Learning • Computer Vision • Natural Language Processing

Actual Location

Padua (PD), Italy

Hometown

Cassino (FR), Italy

Contacts:

alessandro.f.job@gmail.com

+39 320 535 6186



WORK EXPERIENCE

Feb 2024–Oct 2025

AI Research Fellow at Italian Institute of Technology, main activities include: (Roncade (TV))

- Developed methods for spectral unmixing and pigment identification in archaeological frescoes, focusing specifically on hyperspectral data from Pompeii. Designed unsupervised clustering algorithms for remote sensing data.
- Preparation of European project proposals on anomaly detection in cultural heritage, using both supervised and unsupervised machine learning techniques applied to visual data.
- Development of link prediction techniques using Graph Neural Networks to uncover illicit transactions of cultural heritage objects.
- Development of image segmentation pipelines using transformers for ancient cuneiform tablets.

Feb 2023–Jul 2023

Research Internship in Computer Vision at University of Padua

DEGREES

2023

Physics of Data

M.Sc. • Padua 

University of Padua



2020

Physics

B.Sc. • L'Aquila 

University of L'Aquila



TECH SKILLS

Programming: Python, Matlab

Shell: Bash (Linux), PowerShell (Windows), Git Bash, SSH

Database: SQL, PostgreSQL, MongoDB

Versioning: GitHub, GitLab

Frameworks & Tools: Docker, FastAPI, Streamlit, LangChain, AWS

RELEVANT PROJECTS

- End-to-end Development of a Chatbot Application (Backend & Frontend)
 - **Tools:** FastAPI, Streamlit, LangChain, SQLAlchemy, AWS Bedrock
- Porting to AWS of an Agentic Search system for long-form video understanding
 - **Tools:** LangChain, AWS Bedrock
- Advanced clustering techniques using Optimal Transport theory for spectral imaging (*Publishing*)
 - **Tools:** MATLAB, Python, NumPy, JAX, scikit-learn, Matplotlib
- Graph-Based Link Prediction Using GNNs for illicit traffic of artworks (*Publishing*)
 - **Tools:** PyTorch, Torch Geometric, NetworkX, Docker

CERTIFICATES

2025 AWS Certified Data Engineer - Associate (707/1000)

2025 AWS Cloud Practitioner Essentials

2025 MongoDB with Python

2024 English B2

HOBBIES

